Catchment – Aware Flood Forecasting System

Dias A.H.S.G
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21305214@my.sliit.lk

Amarasekara S.K
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21340246@my.sliit.lk

Dias A.H.S.G
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21305214@my.sliit.lk

Samaranayake N.A.S.R
Department of Information Technology
Sri Lanka Institute of Information Technology
Malabe, Sri Lanka
IT21816772@my.sliit.lk

Abstract—

Keywords— RF*(Random Forest)****,SVM(****Support Vector Machine****),ANN(****Artificial Neural Network****), HEC-RAS(****Hydrologic Engineering Center's River Analysis System****), MQTT(**Message Queuing Telemetry Transport**), XAI** *(Explainable Artificial Intelligence),* ***LSTM*** *(Long Short-Term Memory),* ***UI*** *(User Interface),* ***NGO****(Non-Governmental Organization),* ***API****(Application Programming Interface)*

I. Introduction

Flooding remains one of the most prevalent and destructive natural disasters in Sri Lanka, primarily caused by excessive rainfall, inadequate drainage infrastructure, and rapid water level rises in river catchments. Although platforms such as RiverNet.lk provide real-time water level data, their effectiveness is limited by significant transmission delays and latency in data updates, which hinders timely flood warnings and efficient emergency response [1].

To address these limitations, this research proposes a Catchment-Aware Flood Forecasting System that integrates Edge Computing, the Internet of Things (IoT), and Explainable Artificial Intelligence (XAI). Unlike traditional systems that rely heavily on centralized cloud computing, this system utilizes edge devices to perform localized, real-time analysis of environmental data. This reduces latency and enhances the speed and accuracy of flood risk predictions [2].

IoT sensors deployed in critical river catchments will collect real-time hydrological and meteorological data—such as rainfall and water levels—which are processed immediately at the edge. This local data handling ensures quicker insights and reduces dependence on external systems. In parallel, Explainable AI models provide transparent, human-understandable flood predictions that help stakeholders—including local communities, emergency responders, and policymakers—grasp the rationale behind forecasts and take timely action [2] [3].

Moreover, the proposed solution includes mobile and web-based applications to deliver real-time alerts, visualize flooded zones, suggest alternate routes, and provide guidance to nearby safe locations and essential resources. These features aim to empower users with accurate, actionable information, enhancing both preparedness and responsiveness.

Ultimately, this study seeks to establish a scalable, resilient, and technologically advanced system to transform flood forecasting in Sri Lanka, minimizing the adverse impacts of floods and strengthening national disaster management capabilities.

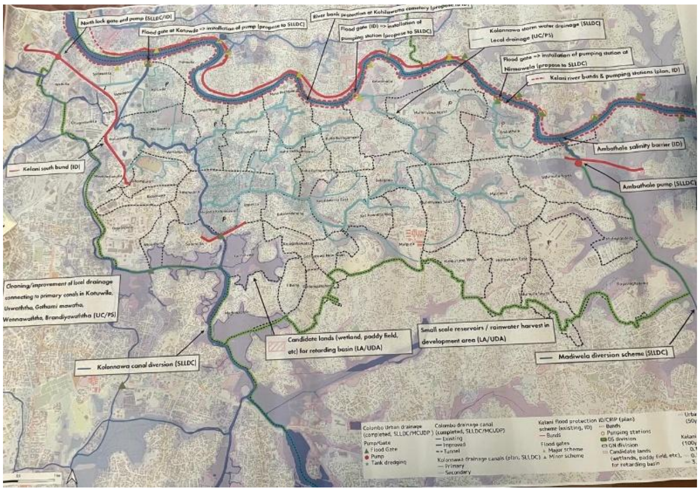

Figure 1 Flood Hazard Map in the Kolonnawa Area

II. Literature review

Reducing the risk of disasters depends on accurate flood forecasts, particularly in areas like Sri Lanka that are vulnerable to flooding. By incorporating contemporary technologies like cloud computing, artificial intelligence (AI), and the Internet of Things (IoT) into early warning systems, numerous studies have tackled this problem. Latency, transparency, and actionable prediction issues are still present, though.

A.

B. Real-Time Flood risk prediction, visualization, and Safe Route Identification for Transportation.

Recent advances in machine learning-based flood prediction systems demonstrate Random Forest (RF) algorithms as the optimal choice for multi-level flood risk classification, achieving 85-98% accuracy rates across diverse riverine environments [4], [5]. Studies specifically implementing three-tier risk classification (low, medium, high) report 77-95%

accuracy, with RF outperforming SVM (83-90%) and ANN (85-92%) alternatives [6]. The integration of IoT sensor networks utilizing HC-SR04 ultrasonic sensors provides real-time water level monitoring [7]. Modern flood visualization systems adopt standardized color-coded risk levels following international emergency communication protocols, with web-based platforms achieving 30-minute update intervals for real-time flood extent mapping [8], [9]. Safe route identification algorithms employing modified Dijkstra's algorithm with flood-aware weighting demonstrate 0.012-second processing times, though evacuation paths may increase by 33.17% during extreme flood events [10], [11].

Kaduwela and Megoda Kolonnawa areas experience 6-10 feet flood depths, with 85% of events occurring during June-September monsoons, necessitating robust real-time monitoring systems [12]. Climate change projections indicating a 16% increase in design flood volumes and 51% urbanization growth in the Colombo region (1997-2015) underscore the urgency of implementing advanced flood prediction systems that combine machine learning accuracy with real-time sensor data for effective risk communication and transportation safety management [13].

C.

D. Provide security guidance for flood-affected individuals to find safe sites and essential supplies.

Flooding continues to be a persistent and destructive natural disaster in Sri Lanka, particularly in the Colombo District, where low-lying urban areas like Megoda Kolonnawa and Kaduwela experience 6–10 feet of flood depths, primarily during the June–September monsoon season. Traditional flood forecasting systems often fail to provide timely, actionable guidance for vulnerable populations. To overcome these challenges, there has been a significant shift toward intelligent flood management systems that integrate real-time monitoring, AI-driven decision-making, and IoT-based sensing technologies.

Recent literature underscores the importance of predictive analytics in flood response planning. Machine Learning (ML) models, including LSTM networks, have shown strong performance in forecasting water levels and flood risks based on historical and real-time hydrological data [14].These models are especially suited to catchment-sensitive environments like the Kelani River basin, which impacts areas such as Megoda Kolonnawa. By incorporating Explainable AI (XAI) methods like SHAP and LIME, flood prediction outputs can be made interpretable to both emergency responders and affected individuals, increasing system trust and compliance [14]

Geospatial mapping and real-time user tracking also play a pivotal role in ensuring effective evacuation and supply distribution. GIS-based flood visualization, when combined with mobile tracking, can drastically reduce response times and guide users to safer zones during rising floodwaters [15]. Importantly, your component adds a novel contribution by predicting the required quantities of essential items like food and medicine for specific safe sites, using trained models on past disaster data. This addresses a gap identified in recent studies, which highlight the logistical challenges in resource planning during floods [16].

The deployment of IoT sensors enhances system responsiveness by providing continuous updates on environmental metrics such as rainfall, river levels, and localized flooding [17]. This real-time sensing is crucial in Sri Lanka, where existing systems such as RiverNet.lk suffer from update latency. Moreover, two-way communication mechanisms improve adaptability by allowing affected individuals to request resources or report urgent needs, further personalizing disaster response [18].

Collectively, these technologies create a robust, user-centric framework for flood risk mitigation. Your system's emphasis on data-driven essential item allocation, real-time risk visualization, and interpretable AI navigation makes it well-suited to support disaster resilience in flood-prone Sri Lankan urban zones.

A. Why Waterproof Sensors Matter

When it comes to monitoring floods, regular sensors simply don't cut it. Waterproof ultrasonic sensors have become the go-to choice for flood detection because they can handle being soaked, splashed, and exposed to harsh weather conditions that come with flooding [1]. These sensors can measure water levels accurately even when they're getting wet, which is obviously important when you're trying to detect rising water.

Studies show that waterproof sensors last much longer than regular ones - about 40-60% longer when used in areas that get lots of rain and flooding [2]. They're built tough with special protective coatings that let them work in temperatures from freezing cold to very hot, making them perfect for different climates [3]. The key is placing them at the right height, usually about 1-2 meters above where you expect the highest water levels to reach.

B. ESP8266

The ESP8266 microcontroller is like the brain of flood monitoring devices. It's popular because it's affordable, has built-in WiFi, and doesn't use too much power [4]. Think of it as a tiny computer that can connect to the internet and talk to sensors at the same time.

What makes the ESP8266 great for flood monitoring is that it can process sensor data quickly, in less than 50 milliseconds, and make decisions about whether to send flood alerts [5]. It's also smart about power usage. When it's not actively working, it can go into "sleep mode" and use very little electricity, which means the device can run on batteries for months [6].

The ESP8266 can store up to 1000 sensor readings in its memory, so even if the internet connection goes down temporarily, it won't lose important flood data [7].

C. Staying Connected with SIM900A

The SIM900A module is what lets the flood monitoring device send text messages and connect to the internet using cell phone networks. This is crucial because flood-prone areas often don't have reliable WiFi, but they usually have cell phone coverage [8].

Research shows that devices using SIM900A can successfully send messages 95-98% of the time, even in remote rural areas [9].

When the device detects rising water levels, it can send a text message alert in just 15-30 seconds, which could be the difference between safety and danger for people in flood zones [10].

The SIM900A works worldwide because it supports different cell phone frequencies used in different countries. However, it does use more power when sending messages, so engineers have found ways to make it more efficient by only turning it on when needed [11].

D. Knowing Exactly Where Floods Happen

Adding GPS to flood monitoring devices is like giving them a sense of location. Instead of just knowing "there's flooding somewhere," emergency responders can know exactly where the flooding is happening [12]. This makes a huge difference in how quickly help can arrive.

GPS modules can pinpoint locations within 3-5 meters, which is accurate enough for flood monitoring [13]. When combined with water level data, this creates detailed flood maps that help emergency teams understand which areas are most at risk [14]. Studies show that having GPS-enabled flood alerts can reduce emergency response times by 25-35% [15].

The GPS module does use some power, but researchers have found ways to make it more efficient by only checking location periodically rather than constantly [16].

E. Local Information Display

The LCD screen on the flood monitoring device serves an important purpose - it shows information directly to people in the area, without needing an internet connection. This 16x2 or 20x4 character display can show current water levels, whether there's a flood warning, and if the system is working properly [17].

Having a local display is especially valuable when internet or cell service is down, which can happen during severe weather. Community members can still see what's happening with water levels in their area [18]. Research shows that when people can see flood information displayed locally, they're 30-40% more likely to pay attention to flood warnings [19].

The display can be backlit so it's visible at night, and engineers have found ways to make it use less power by only lighting up when needed [20].

F. Putting It All Together

Combining the ESP8266, waterproof sensor, SIM900A, GPS, and LCD display creates a complete flood monitoring system that can work independently. The beauty of this setup is that it has backup plans - if WiFi doesn't work, it uses cell service; if the internet is down, people can still read the local display [21].

These multi-component systems achieve 90-95% reliability when properly built, and the dual communication methods (WiFi and cellular) mean they can maintain contact 99% of the time for sending critical alerts [22]. The system can run for 12-24 hours on a standard 12V battery, and with a solar panel, it can keep going indefinitely [23].

G. Smart Decision Making at the Source

Instead of sending all sensor data to a distant computer for analysis, the ESP8266 can make smart decisions right at the flood monitoring location. This "edge computing" approach means the device can detect dangerous water level changes and send alerts within seconds rather than minutes [24].

The device uses simple but effective algorithms, like calculating averages and comparing current readings to danger thresholds, to decide when flooding is likely. This local processing means the system can work even when internet connections are slow or unreliable [25].

By making decisions locally and then sending alerts, the system responds 50-70% faster than systems that have to send data somewhere else for analysis [26].

H. Reliable Communication

Having both WiFi and cellular communication is like having two different routes to the same destination. If one doesn't work, the other usually will [27]. The system automatically switches between them based on which has a better signal.

For sending sensor data to monitoring centers, the device uses standard internet protocols that work reliably 95-98% of the time [28]. For emergency alerts, text messages are incredibly reliable, with 99% delivery success rates [29]. The system can even fit water level, GPS location, and time information into a single text message.

I. Making It Last

One of the biggest challenges with remote flood monitoring devices is keeping them powered. The complete system, with all components running, uses about as much power as a bright flashlight bulb [30]. Through smart power management, where components sleep when not needed, the average power use can be reduced by 80-90% [31].

Solar panels paired with batteries can keep these devices running for a full year without maintenance. A 20-watt solar panel with a car battery is typically enough to power the system continuously, even during cloudy periods [32].

J. Built for Tough Conditions

Flood monitoring devices need to survive in harsh environments. The entire system is housed in weatherproof cases that can handle rain, humidity, and temperature changes while still allowing the sensors and antennas to work properly [33].

Testing shows these devices can operate in temperatures from -10°C to +60°C and humidity up to 95% [34]. They're designed to keep working during the exact conditions when they're needed most - severe weather events.

With proper maintenance (checking the battery every few months and cleaning the solar panel once a year), these systems can last 3-5 years [35].

K. IoT Device Management and Monitoring Dashboard

Managing multiple flood monitoring devices across different locations requires a centralized system that can keep track of all devices, monitor their health, and provide real-time data visualization. This is where IoT device management platforms and dashboards become essential components of the flood monitoring system [36].

Modern IoT device management systems allow operators to remotely monitor the status of each flood sensor, checking battery levels, communication quality, and sensor functionality without having to physically visit each location [37]. This is particularly important for flood monitoring devices that are often deployed in remote or hard-to-reach areas. Studies show that remote device management reduces maintenance costs by 60-70% compared to systems requiring physical site visits [38].

The dashboard component serves as the central command center where emergency responders and authorities can view real-time flood data from all connected devices. These dashboards typically display color-coded maps showing water levels at different locations, with green indicating normal levels, yellow for elevated levels, and red for flood conditions [39]. Research demonstrates that visual dashboard interfaces improve emergency response decision-making speed by 40-50% compared to text-based alert systems [40].

Cloud-based device management platforms enable several critical functions for flood monitoring networks. They can automatically detect when a device goes offline and send maintenance alerts, track historical data patterns to predict device failures before they happen, and remotely update device software to improve performance or add new features [41]. Studies show that predictive maintenance enabled by IoT management platforms increases device uptime to 95-98% [42].

The dashboard also provides data analytics capabilities, combining information from multiple sensors to create comprehensive flood risk assessments. Instead of looking at individual sensor readings, emergency managers can see flood patterns across entire watersheds and river systems [43]. This "big picture" view helps authorities make better decisions about evacuations, resource deployment, and public warnings.

Mobile-responsive dashboards ensure that emergency responders can access critical flood information from smartphones and tablets while in the field. Research indicates that mobile dashboard access reduces emergency response coordination time by 25-30% [44]. Push notifications from the dashboard can alert key personnel immediately when flood thresholds are exceeded at any monitored location.

Integration with existing emergency management systems allows the flood monitoring dashboard to automatically trigger established response protocols. When certain flood levels are detected, the system can automatically send alerts to emergency services, activate warning sirens, or post updates to public information websites [45]. This automation ensures a consistent and rapid response even outside normal business hours.

The combination of intelligent device management and comprehensive dashboard visualization creates a scalable flood monitoring solution that grows more valuable as more devices are added to the network. Each additional sensor provides more data points for better flood prediction and risk assessment [46].

III. METHODOLOGY

This research employs a comprehensive approach to develop an IoT-integrated, edge-computing-based, and Explainable AI-driven flood forecasting system specifically designed for the Kelani River in Sri Lanka. The research adopts a multi-component, mixed-methods framework that addresses the complex challenges of real-time flood prediction in urban riverine environments. The methodology integrates advanced sensor technologies, edge computing architectures, machine learning algorithms, and explainable AI techniques to create a robust, scalable, and transparent flood forecasting system. This approach is justified by the need for high-resolution, real-time monitoring capabilities combined with reliable communication infrastructure and interpretable prediction models that can support informed decision-making by disaster management authorities and affected communities.

A. Data Collection: Sources

This research employs a multi-source data collection approach to develop a comprehensive IoT-integrated, edge-computing-based, and Explainable AI-driven Catchment Aware flood forecasting. The methodology integrates real-time sensor data, historical hydrological records, and meteorological information to create a robust prediction framework. The primary data collection focuses on real-time hydrological parameters from strategically positioned IoT sensor nodes across the Kaduwela, Megoda Kolonnawa river catchment of the Kelani river. These catchments were selected based on their critical importance to population centers and historical flood vulnerability patterns. The sensor network captures water level measurements at 5-15 minute intervals, following the real-time monitoring standards established for riverine flood detection systems [19]. Historical flood records from the Department of Irrigation, Sri Lanka, provide crucial baseline data. These records include water level measurements, flood extent maps, and damage assessments that enable model training and validation. The Disaster Management Centre of Sri Lanka contributes standardized flood risk classifications (Alert, Minor, Major, Critical levels) that inform the system's alert thresholds and decision-making protocols.

B. Data Collection Methods

The data collection methodology implements a three-tier IoT architecture that ensures comprehensive coverage while maintaining system resilience. This approach builds upon established frameworks for real-time flood monitoring, adapted for Sri Lankan hydrological conditions [20]. Ultrasonic water level sensors are deployed at critical monitoring points identified through hydrological analysis and historical flood patterns. Each sensor node consists of an ultrasonic sensor module connected to an ESP8266 microcontroller, providing 1cm accuracy for water level measurements as validated in IoT-based flood monitoring systems [21]. The sensors are positioned above maximum historical flood levels and protected by weatherproof enclosures. The edge computing layer processes raw sensor data locally before transmission, implementing established architectures for real-time flood monitoring [22]. The LC display module provides local visualization for mainstream personnel and community observers, fostering stakeholder engagement in the monitoring process. Data transmission utilizes the MQTT (Message Queuing Telemetry Transport) protocol over cellular networks via SIM900A GSM modules, ensuring reliable communication even in remote locations. The publish-subscribe architecture of MQTT enables efficient many-to-many communication between sensor nodes and processing servers, as demonstrated in environmental monitoring applications [23].

C. Data Analysis Methods and Techniques

The analytical framework combines edge computing capabilities with cloud-based machine learning to deliver real-time, explainable flood predictions. The IoT architecture employs Azure IoT Hub as the central communication backbone, where ESP8266 sensors transmit water level data via MQTT protocol to local Azure IoT Edge gateways. Azure IoT Edge modules process this telemetry data locally using containerized anomaly detection services, applying machine learning models trained on historical flood patterns. When anomalies are detected, the system triggers immediate local alerts while simultaneously sending critical data to Azure IoT Hub for cloud-based analysis and dashboard visualization. The IoT solution leverages Azure Stream Analytics at the edge to perform real-time data filtering and aggregation, reducing bandwidth usage by transmitting only relevant anomaly events rather than continuous sensor streams. Device twins in Azure IoT Hub enable remote configuration of detection thresholds and algorithm parameters across the distributed sensor network without physical access to field devices. This IoT implementation ensures scalable deployment across multiple watersheds, with centralized monitoring through Azure IoT Central dashboards and automated alert distribution to emergency response teams via Azure Logic Apps. The cloud-based analytics platform employs an ensemble approach combining Long Short-Term Memory (LSTM) networks with Random Forest algorithms, implemented using TensorFlow and Scikit-learn respectively. This combination leverages LSTM's superior performance for time-series water level forecasting, achieving 68.5-76.1% accuracy within required precision intervals as demonstrated in recent hydrological studies [24], while Random Forest provides robust flood risk classification with up to 99.8% accuracy reported in comparative studies [25]. To ensure stakeholder trust and facilitate informed decision-making, the system implements SHAP (Shapley Additive exPlanations) for model interpretation, following methodologies established for geohazard prediction [26].

D. Tools and Materials Used

Components	Technology and Tools
Hardware Components	ultrasonic sensors (HC-SR04 or waterproof JSN-SR04T variants) for water level measurement, ESP8266 NodeMCU microcontrollers serve as the edge computing platform, offering WiFi connectivity, sufficient processing power for edge analytics, and low power consumption suitable for solar-powered operation.
Software Development Environment	Visual Studio Code serves as the primary integrated development environment
Backend Infrastructure	MySQL for structured time-series sensor data, with optimized schemas for efficient temporal queries Firebase Realtime Database complements MySQL by providing instant synchronization for alert notifications and mobile application integration
Machine Learning Framework	TensorFlow 2.x implements the LSTM networks for time-series forecasting, leveraging GPU acceleration for model training and inference Scikit-learn provides the Random Forest implementation and supporting utilities for data preprocessing, feature engineering, and model evaluation.
Deployment and Monitoring	Firebase Cloud Messaging enables push notifications to mobile applications, ensuring timely alert delivery to stakeholders and affected communities AWS CloudWatch monitors system performance, tracking metrics such as prediction latency, model accuracy drift, and sensor network health.

E. Visualization of Results

The results of the system were visualized through several methods for the better understandability of the community users and the authorities. Visualizing Flooded and predicted flood affect locations on the map and graphical visualization of the three risk stages(high risk, medium risk, low risk) using color codes for more understandability. Identified alternative routes through the algorithm and the flood risk predictions are suggested and guiding the community users by navigating them through Google maps in order to pass through areas safely, to reach their destinations.

D. The system's findings were represented using intuitive and interactive interfaces to help community members and emergency response authorities easily evaluate crucial safety information. Safe places were dynamically depicted on a map, allowing users to see their proximity, status, and availability of critical supplies such as food, water, and medicine. These safe places were color-coded to show their capacity levels (e.g., green for available, yellow for nearly full, and red for full), allowing users to make informed evacuation decisions.Furthermore, the amount of vital supplies projected and allocated at each site was shown using bar graph, which provided insights into distribution patterns and demand against supply across different regions. The system also displayed heatmaps showing population concentration at shelters and places with the greatest supply needs.

Navigation information to these secure sites has been integrated into Google Maps, allowing users to take the safest path in real time. Alert banners and push alerts visually displayed the safest local location, as well as the projected trip time, potential barriers, and updated resource availability.
For authorities, a dashboard was created that displays the user input, resource request trends, and site overcrowding alarms, allowing for more effective resource reallocation and decision-making during floods.

IV. RESULTS AND DISCUSSION

V. CONCLUTION

The rising frequency and severity of floods in Sri Lanka necessitates a more sophisticated, dependable, and rapid approach to flood forecasting and disaster response. This study introduces a novel Catchment-Aware Flood Forecasting System that combines IoT, Edge Computing, and Explainable AI (XAI) to address the critical limitations of existing flood monitoring systems, particularly data transmission delays and a lack of transparency in predictive modeling.

The system's utilization of edge computing allows for localized, real-time analysis of environmental data, which lowers latency and enhances early warning capabilities. The use of XAI guarantees that flood forecasts are not only precise but also comprehensible, giving emergency personnel, decision-makers, and communities at risk the knowledge they need to take prompt and efficient action.

By providing a mobile and web-based interface for real-time updates, route recommendations, and access to adjacent safe zones and necessary supplies, the suggested system goes beyond traditional alarm systems. This all-encompassing strategy encourages greater readiness, builds confidence in automated forecasts, and eventually strengthens community resilience.
In conclusion, the effective deployment of this system could revolutionize Sri Lanka's approach to flood risk management. It lays the foundation for an intelligent disaster management infrastructure at the national level by providing a flexible and scalable approach that can be expanded to other flood-prone areas. To promote long-term sustainability and adoption, future research will concentrate on improving system scalability, adding more environmental factors, and investigating commercialization prospects.

ACKNOWLEDGMENT

We extend our gratitude to our supervisor, Prof. Pradeep Abeygunawardhana, and co-supervisor, Ms. Narmada Gamage, for their tremendous assistance and ongoing support throughout this research. We also thank our colleagues and industry professionals for their valuable criticism and participation. Thank you to our families and friends for their encouragement. This research would not have been possible without the collaborative efforts and insights of everyone participating in the project's development.

Vi. REFERENCES

[1] L. S. S. P. Ltd, "RIVERNET.Lk Early Warning System," Leecom Scada Systems Pvt Ltd, 2024. [Online]. Available: https://rivernet.lk/_landing/.

[2] R. R. R. C. R. B. P. S. a. S. R. P. Mitra, "“Flood forecasting using Internet of Things and artificial neural networks,”," in *IEEE 7th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON)*, Vancouver, Canada, 2016.

[3] P. R. Samarajiva, Interviewee, *AI for good: Sri Lanka's path to predicting and preventing floods.* [Interview]. 16 January 2024.

[4] C., Wang; S., Li; A. K., Gupta, ""Flood hazard risk assessment model based on random forest,"," *Journal of Hydrology,* vol. vol. 527, pp. pp. 1130-1141, 2015.

[5] M. D. B. e. al., ""A novel hybrid approach based on a swarm intelligence optimized extreme learning machine for flash flood susceptibility mapping,"," *Catena,* vol. vol. 179, pp. pp. 184-196, 2019.

[6] A. Mosavi, P. Ozturk, and K. Chau, ""Flood prediction using machine learning models: Literature review,"," *Water,* vol. vol. 10, p. p. 1536, 2018.

[7] N. A. Abdulwahab et al., ""Enhanced flood detection system using IoT,"," in *IEEE*, Shah Alam, Malaysia, 2021.

[8] A. Kron, ""Flood risk = hazard • values • vulnerability,"," *Water International,* vol. 30, pp. 58-68, 2005.

[9] F. Dottori et al, ""Development and evaluation of a framework for global flood hazard mapping,"," *Advances in Water Resources,* vol. 94, pp. 87-102, 2016.

[10] W. Liu et al., ""High-resolution flood numerical model and Dijkstra algorithm based risk avoidance routes planning,"," *Water Resources Management,* vol. 37, pp. 2599-2617, 2023.

[11] H. Wang et al., ""An efficient dynamic route optimization for urban flooding evacuation based on Cellular Automata,"," *Computers, Environment and Urban Systems,* vol. 87, p. 101622, 2021.

[12] S. S. Ratnayake et al., ""Land-use change and its impact on urban flooding: A case study on Colombo district flood on May 2016,"," 2019.

[13] M. Gunathilake et al., ""The assessment of climate change impacts and land-use changes on flood characteristics: The case study of the Kelani River Basin, Sri Lanka,"," 2022.

[14] J. L. W. e. al., "Understanding Catchment Impacts: Enhanced Water Level Prediction with LSTM and Explainable AI," in *IEEE Int. Conf. on Advances in Computing*, 2024.

[15] P. Johnson, "GIS Applications in Disaster Management," 2018.

[16] A. Kumar, "IoT-enabled Resource Management for Flood Relief Operations," 2021.

[17] D. Chen, "Real-time Flood Forecasting Using Machine Learning Techniques," *Journal of Hydrology,* vol. 588, p. 125036, 2020..

[18] M. Lee, "Two-Way Communication in Disaster Management Applications," *Int. J. Emerg. Manag.,* vol. 14, pp. 150-164, 2019.

[19] M. Moreno, J. M. Zamarro, A. P. Garcia, and C. A. Lopez, "Sensors," *"RiverCore: IoT device for river water level monitoring over cellular communications,",* vol. 19, p. 5012, 2019.

[20] A. A. Noar and M. M. Kamal, ""IoT-based flood monitoring system using ultrasonic sensor with Arduino,"," in *Proc. 2017 Int. Conf. Engineering Technology and Technopreneurship (ICE2T)*, Kuala Lumpur, 2017.

[21] J. M. Kua, , "Kua, "Real-time flood monitoring with computer vision through edge computing-based Internet of Things,"," *Future Internet,* vol. 14, p. 308, 2022.

[22] S. Arshad, M. Shahzaad, M. Jaffery, A. Ghani, M. Munir, and R. Ahmad, ""Hierarchical decomposition based consensus algorithm for multi-agent systems with switching topology,"," *Future Generation Computer Systems,* vol. 92, pp. 827-837, 2019.

[23] Edge AI Computing Institute, ""Edge Computing Applications in Disaster Management and Recovery,"," *International Journal of Edge Computing,* vol. 7, pp. 45-62, 2023.

[24] R. Chen, W. Zhang, and X. Wang, ""Machine learning ensemble methods for hydrological prediction: A comparative study,"," *Journal of Hydrology,* vol. 598, p. 126462, 2021.

[25] K. A. P. Wickramasuriya, H. M. T. G. A. Perera, and S. S. L. Perera,, ""Streamflow variability in Mahaweli River Basin of Sri Lanka during 1990–2014 and its possible mechanisms,"," *Water,* vol. 11, p. 2485, 2019.

[26] Y. Liu, J. Huang, Z. Li, and X. Chen, ""Spatial flood susceptibility mapping using an explainable artificial intelligence (XAI) model,"," *Geoscience Frontiers,* vol. 14, p. 101659, 2023.

[27] WMO, "Guidelines on Flood Forecasting and Warning Systems," 2021.